

Day 2 — KG-Adapter: Parameter-Efficient Fine-Tuning for Knowledge Graph Integration

LLM Research Log | Souvik Pal

Mentor: Dinesh Sir • Research Area: Large Language Models (LLMs)

KG-Adapter: Enabling Knowledge Graph Integration in LLMs through Parameter-Efficient Fine-Tuning (PEFT)

Objective

To understand how Knowledge Graph knowledge can be integrated into an LLM's internal layers — rather than only supplied externally through prompts — using a lightweight, trainable adapter module.

The Problem

Beyond hallucination, LLMs face several related limitations: limited domain expertise, knowledge that goes stale without retraining, disagreement between the model's internal memory and external facts, heavy dependence on prompt quality, and poor explainability of why an answer was generated.

Existing approaches each fall short in a different way:

Method	Approach	Key Problem
Traditional LLM	Uses only trained memory	Hallucinations, outdated knowledge
Prompt Engineering	Injects KG facts into the prompt	Output quality depends entirely on prompt design
RAG	Retrieves documents at inference	Ignores graph structure; no real reasoning over relations
KG Prompting	Puts KG triples directly in the prompt	Prompt becomes too long; still prompt-dependent
Full Fine-Tuning	Retrains the entire model on KG data	Extremely expensive; risks forgetting prior knowledge

Methodology

Core idea: instead of placing KG facts inside the prompt text, KG-Adapter attaches a small trainable module directly inside the frozen LLM. The pipeline is: Knowledge Graph → Graph Encoder (GNN-based) → KG Adapter (lightweight, trainable) → LLM Decoder (frozen) → Generated Response.

Two complementary encoding strategies make this work:

- **Node-centered encoding** — each entity in the KG (a person, place, concept) is converted into a vector representation that the adapter can use
- **Relation-centered encoding** — the edges between entities (e.g. "Harrier → has temperament → Friendly") are encoded separately and combined with node vectors, since knowing an entity alone is incomplete without knowing how it relates to others

This is a direct application of **Parameter-Efficient Fine-Tuning (PEFT)**: rather than updating all of a model's weights (expensive, memory-heavy, and prone to catastrophic forgetting), only the small adapter module is trained while the base LLM stays frozen. In practice, this meant training only **28 million parameters out of a 7-billion-parameter model — just 0.4% of the total.**

Experimental Setup

- **Base models tested:** LLaMA (7B), Vicuna (7B), ChatGLM
- **Benchmarks:** WebQSP, CWQ (Complex Web Questions), GrailQA
- **Tasks:** Knowledge Graph QA, entity linking, relation reasoning
- **Metrics:** F1 Score, Hits@1, Accuracy, Exact Match

Results

KG-Adapter outperformed all baseline methods across the KGQA benchmarks while training only ~0.4% of total model parameters — showing that learning structured knowledge through a graph encoder is more effective than simply retrieving it via prompts or RAG.

My Assessment — Limitations

- Performance depends entirely on the quality of the input KG — errors or gaps in the graph propagate directly into model output
- The adapter has no mechanism to correct wrong KG data; it learns whatever the graph contains, including its errors
- Building a good domain-specific KG is itself expensive and expert-intensive
- Adapter performance is still bounded by the base LLM's own capability
- Real-time KG updates are difficult — dynamic changes require re-training or re-initializing the adapter
- Evaluation so far is mostly limited to KGQA-style tasks, not the full diversity of real-world NLP use cases

My Suggestions for Future Improvement

- Automatic KG updating as new facts emerge
- Confidence scoring before the model commits to an answer
- Multi-KG support — integrating multiple domain graphs at once
- Adaptive adapter selection per domain or query type
- Visual explanation of the graph reasoning path behind an answer

- Self-correction — re-verifying the answer against the KG post-generation
- Ranking KG facts by source trustworthiness
- Reasoning-path memory — reusing successful inference paths across queries

Conclusion

Traditional LLMs rely mostly on learned memory, which drives hallucination and factual drift. KG-Adapter offers a more structural fix: injecting knowledge directly into model internals via a lightweight, PEFT-trained module, achieving strong accuracy gains at a fraction of the training cost of full fine-tuning. The key distinction from Day 1's approaches is that KG-Adapter learns knowledge, rather than just retrieving it — a meaningfully different strategy from prompt-based or RAG-based methods.

What I Personally Took Away

- The distinction between adding knowledge outside the model (prompting) versus inside it (adapters)
- How Graph Neural Networks (GNNs) are used to turn graph structure into vectors usable by a language model
- Why lightweight fine-tuning (PEFT) matters for making serious LLM research feasible without massive GPU infrastructure
- How to read a research paper critically: identifying not just what it claims, but its limitations and what I would improve

Personal Reflection

For Day 2, Sir gave us resources related to KG-Adapter, PEFT, GNN, and a few other topics. After going through the research paper, it looked quite complex, so I used ChatGPT to help make sense of the harder parts. But later, when I discussed it with Dinesh Sir, I realized that ChatGPT can also give wrong answers sometimes. That was an important lesson I learned that day — not to blindly trust an AI's explanation without cross-checking it.